Vietnamese GraphRAG: Knowledge Graph Question Answering over Wikipedia with Local SLM + Neo4j

Presenter: [Student Name] — [Student ID]

FPT University

June 2026

FPT UNIVERSITY

Agenda

Problem & Research Gap

Three Critical Gaps in Vietnamese QA

- **Single-hop only** — no multi-hop graph reasoning in existing systems
- **Hallucination** — 57% unfaithful citations (Liu et al., 2024)
- **API dependency** — cost, latency, data sovereignty concerns

No fully-local Vietnamese KGQA system exists for on-premise deployment with multi-hop reasoning.

Research Objectives

	Objective	Deliverable	Metric
O1	Local GraphRAG system	End-to-end pipeline	Latency < 3s
O2	Open multi-hop dataset	ViWiki-MHR $\sim$36K	6 reasoning types
O3	Citation groundedness	CyVer + passage IDs	Faithfulness
O4	Standard benchmarks	ViQuAD 2.0 eval	F1 > 0.40

Constraint: consumer hardware only — 16GB RAM, 8GB VRAM, no external APIs

Related Work & Positioning

System	Approach	Gap We Address
GraphRAG (Edge et al., 2024)	Community summaries + global search	Enterprise-scale, not local
StepChain (2025)	Question decomposition + BFS on KG	English-only, no Vietnamese NER
HisGraphRAG (PACLIC 2025)	GraphRAG for Vietnamese history	Domain-specific, textbook only
CyVerACT (Androna et al., 2026)	Deterministic Cypher validation	No agent loop, no Vietnamese
GFM-RAG (2025)	Graph Foundation Model 8M params	Requires pre-training
ReflectiveRAG (Verma et al., 2026)	Sufficiency checks + re-retrieval	No KG, English-only

Our position: Local SLM + typed Vietnamese KG + deterministic tool-calling + open dataset

Key Contributions

1. GraphRAG System

- Typed Neo4j KG + 3 pluggable NER backends
- Hybrid retrieval + cross-encoder reranking
- ReAct agent with 4 deterministic tools

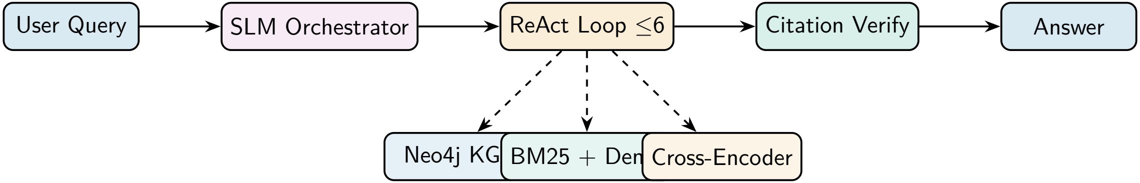
2. ViWiki-MHR Dataset

- ~36K multi-hop QA, 6 reasoning types
- Broken-link adversarial unanswerables
- 3-stage QC (grounding, NLI, dedup)

3. Local SLM Pipeline

Qwen2.5-7B (4-bit NF4) · Text2Cypher + CyVer · 8GB VRAM

System Architecture



- **Fully local** — no external API calls; 16GB RAM, 8GB VRAM (RTX 3060/4060)
- **Model:** Qwen2.5-7B-Instruct, 4-bit NF4 (∼5.5GB VRAM)
- **Reranker:** BAAI/bge-reranker-v2-m3 (multilingual cross-encoder)

Knowledge Graph Construction

Typed Schema

```
(:Page) -[:HAS_CHUNK]-> (:Chunk)
(:Chunk)-[:MENTIONS_PERSON]->(:Person)
(:Chunk)-[:MENTIONS_ORG]->(:Organization)
(:Chunk)-[:MENTIONS_LOC]->(:Location)
(:Chunk)-[:MENTIONS_WORK]->(:Work)
(:Page) -[:LINKS_TO]-> (:Page)
```

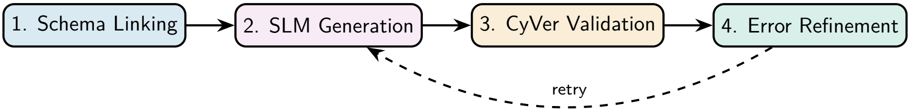
Current scale: 998 pages, 19K chunks, 3.3K entities

3 Pluggable NER Backends

- **Simple** — regex heuristic + keyword type classification
- **Underthesea** — BIO tagging, offline Vietnamese NER
- **PhoNLP** — VnCoreNLP word segmentation + PhoNLP NER

All backends → `(name, type)` tuples → typed graph labels

Text2Cypher Translation Pipeline



Example: "Tác phẩm 'Tắt đèn' được sáng tác bởi nhà văn sinh năm bao nhiêu?"

2-hop Vietnamese → Cypher

```
MATCH (tp:TacPham {ten: "Tat den"})<-[:SANG_TAC]-(nv:NhaVan)
RETURN nv.nam_sinh
```

Inspired by CyVerACT (Androna et al., 2026) and Ozsoy et al. (2025, 2026)

Hybrid Retrieval with Cross-Encoder Reranking

1. **Schema Linking** — BM25/TF-IDF prunes KG schema to relevant labels/relations
2. **Cypher Generation** — SLM produces read-only Cypher from query + pruned schema
3. **Deterministic Validation** — syntax check, write-keyword blocklist, alias verification
4. **Hybrid Fallback** — on invalid Cypher $\rightarrow$ BM25 + dense vector retrieval
5. **Cross-Encoder Reranking** — BAAI/bge-reranker-v2-m3 scores top-k

Why Hybrid?

Graph-only degrades under KG sparsity (HybridRAG, 2026). Text fallback ensures coverage.

Reranking Impact

Cross-encoder reranking: +15.5% accuracy (literature benchmark)

Tool-Constrained ReAct Agent

4 Deterministic Tools

- `kg_schema()` — cached JSON schema
- `kg_query(cypher)` — execute & return rows/errors
- `text_search(q, k)` — hybrid BM25 + dense
- `get_passage(id)` — raw paragraph text

Agent Loop (pseudocode)

```
for i in 1..6:
  thought = SLM(obs)
  action = parse_tool(thought)
  obs = execute(action)
  if sufficient(obs): break
answer = cite(obs, passage_ids)
```

Sufficiency gating: empty KG → text_search or abstain. Aligned with ReflectiveRAG, C2RAG.

Local SLM & Fine-tuning Strategy

Runtime Configuration

Parameter	Value
Base Model	Qwen2.5-7B-Instruct
Quantization	4-bit NF4
Loading	Lazy, cached in memory
VRAM	$\sim$5.5GB
Interfaces	`generate()` + `chat()`

Fine-tuning Plan

- **QLoRA** — NF4, double quant, LoRA rank 32, all linear layers, 10–15K pairs
- **DPO Alignment** — JSON tool-call compliance, schema correctness
- **Target:** >95% executable Cypher

Refs: QLoRA (Dettmers, 2023), DPO (Rafailov, 2023)

Data Sources: ViQuAD 2.0 + ViWiki-MHR

UIT-ViQuAD 2.0 (External)

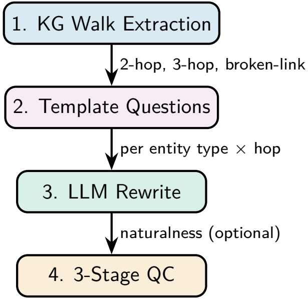
- **39.5K** QA pairs, 174 Wikipedia articles
- Standard Vietnamese MRC benchmark (peer-reviewed)
- ~12K unanswerable questions (SQuAD 2.0 style)

ViWiki-MHR (Ours, CC-BY-SA)

- ~36K multi-hop QA, 6 reasoning types
- Broken-link adversarial unanswerables (first VN)
- Gold passage IDs + executable Cypher queries

Component	**Size**	**Hops**	**Source**
Single-hop answerable	~23K	1	UIT-ViQuAD 2.0
Single-hop unanswerable	~5K	1	ViQuAD 2.0 adversarial
Multi-hop answerable	~7K	2–3	KG walks + LLM rewrite
Adversarial multi-hop	~1K	2–3	Broken-link (first VN)

ViWiki-MHR Generation Pipeline



Reasoning taxonomy: lookup, bridge, comparison, intersection, temporal, fan-out

QC: well-formedness → grounding match → deduplication

Adversarial unanswerables inspired by BRINK (Zhou et al., 2026) — break one link in valid KG path

ViQuAD 2.0 Integration Strategy

Integration Pipeline

1. HuggingFace load + format conversion
2. Dedup (title + text hash)
3. Ingest new passages into KG
4. End-to-end eval (validation split)

Why ViQuAD 2.0?

- **Objectivity** — external, not self-generated
- **Comparability** — CafeBERT: 77.5% EM
- **Abstain testing** — 12K impossible Qs
- **Generative gap** — Token F1 measurement

Success Criteria

Token F1 > 0.40 | Context Hit Rate > 0.60 | Abstain Accuracy > 0.60

Evaluation Framework & Results

Metrics Suite

Category	Metrics
Retrieval	Hit Rate, MRR
Answer	Token F1, EM
Robustness	Abstain Accuracy
Efficiency	Avg Latency
Groundedness	Faithfulness

Baselines

- B1 Vector-only (dense retrieval)
- B2 BM25-only (sparse retrieval)
- B3 Graph-only (Cypher traversal)
- Ours Hybrid (graph + text + reranking)

Results (Placeholder)

Awaiting full evaluation run — targets: F1 $>$ 0.40, Hit Rate $>$ 0.60, Latency $<$ 3s

Live Demo

Sample Multi-hop Query

Q: "Ai sáng lập tổ chức có trụ sở tại Hà Nội?"

Reasoning: Location(Hà Nội) ←[:TRỤ_SỞ]— Organization(X) ←[:SÁNG_LẬP]— Person(?)

Agent trace:


```
{"tool": "kg_query",
 "cypher": "MATCH (l:Location{name:'Hà Nội'})<-[:TRU_SO]-(o:Org)...",
 "result": [{"person": "Nguyễn Văn A", "passage_id": "chunk-0042"}],
 "citations": ["chunk-0042"]}
```

References

- Edge et al. (2024). From Local to Global: A Graph RAG Approach. *Microsoft Research*.
- Androna et al. (2026). CyVerACT: Deterministic Cypher Validation. *arXiv*.
- Ozsoy et al. (2025). Text2Cypher: Schema-Conditioned Generation. *Neo4j Research*.
- Dettmers et al. (2023). QLoRA: Efficient Finetuning of Quantized LLMs. *NeurIPS*.
- Zhou et al. (2026). BRINK: Broken-Link Adversarial Unanswerables. *arXiv*.
- Nguyen et al. (2025). HisGraphRAG. *PACLIC*.
- Nguyen et al. (2022). UIT-ViQuAD 2.0. *UIT-NLP*.
- Liu et al. (2024). Lost in the Middle: How LLMs Use Long Contexts. *TACL*.
- Verma et al. (2026). ReflectiveRAG: Sufficiency-Aware Retrieval. *arXiv*.
- Choudhary et al. (2026). HybridRAG: KG-Augmented Retrieval. *arXiv*.

Conclusion & Future Work

Achieved

- End-to-end local GraphRAG
- Typed Vietnamese KG (998p, 19K chunks)
- ReAct agent + citation tracking
- Dataset pipeline + ViQuAD2
- Multi-metric evaluation framework

Future Directions

- QLoRA fine-tuning (Text2Cypher)
- DPO alignment (tool compliance)
- Community detection (Louvain)
- Query decomposition (StepChain)
- Scale to 5000+ pages

Thank You

Questions & Discussion

Vietnamese GraphRAG over Wikipedia + Neo4j

Local-first · Multi-hop · Citation-grounded

Naive RAG vs. GraphRAG vs. Ours

Feature	Naive RAG	GraphRAG	Ours
Data repr.	Flat chunks	Graph + communities	Typed KG + text
Multi-hop	Poor	Excellent	Excellent + hybrid
Context	Large (raw)	Small (sub-graphs)	Small (Cypher)
Hallucination	None	Structural	CyVer + citations
Cost	Low	High (Leiden)	Medium (consumer)

Broken-Link Adversarial Construction

Valid vs. Broken Path

Valid: (Ngô Tất Tố) —[:SÁNG_TÁC]→ (Tắt đèn) —[:BỐ_CẢNH]→ (Cẩm Giàng)

Broken: (Ngô Tất Tố) —[:SÁNG_TÁC]→ (Tắt đèn) —[:BỐ_CẢNH]→ [Fabricated: Hải Phòng]

- Surface question coherent up to hop 2, fails at hop 3
- Forces agent to verify each path step programmatically
- System must output "I don't know" if link fails
- First Vietnamese benchmark for multi-hop unanswerability

QLoRA Training Configuration

Parameter	Value
Bits	4-bit NormalFloat (NF4)
Double Quantization	Enabled
Target Modules	All linear (q, k, v, o, gate, up, down)
LoRA Rank (r)	32
LoRA Alpha (α)	64
Optimizer	Paged AdamW
Training Data	10–15K Text2Cypher pairs
Evaluation	500 human-verified held-out pairs

Task 1: Vietnamese Text2Cypher (primary)
Task 2: Agent action format conformance (secondary)